The role of demographic feedback on the evolution of developmental windows

Part I
Main Manuscript

1 Introduction

There is an ongoing mass extinction event and, as many of the relevant anthropogenic stressors continue to worsen—not least but not limited to rising global temperatures and the consequent risks of climate system breakdown (CITE)—biological models have an important job to do in predicting the demographic and adaptive responses of species and ecosystems to these changing conditions. There remains much we can do, in principle, to improve these models and so generate more robust and informative guidance for policy makers. For instance, models of demographic change have yet to synthesise reports from the evolutionary theory literature of the great speeds (within ecological timescales) at which adaptation can occur and of the important role of plasticity and agency of individuals in directing this adaptation (CITE). Benton (2012) (CITE), shows how the complexities of real populations resulting from individual-level genetic variation and plasticity can play a strong role in population responses. We need more models like this, elucidating what variables are important in population dynamics, in order to develop appropriate theory, generate accurate empirical predictions, and present robust and effective policy guidance (CITE). Research within behavioural ecology has increasingly focused on the roles of uncertainty, state-dependent decision making, and learning; with implications for the evolution of various behavioural and developmental strategies (CITE CITE CITE), but their roles in population dynamics and the implications for us remain largely speculative (CITE).

1.1 Main question with examples

The main question is, given the predictability and severity of food limitations in an organism's environment, over what length of time do individuals invest in developing higher reproductive quality at an opportunity cost to investing in growth to exploit periods of abundant food?

Developmental windows are periods in development where sensitivity to cues is heightened.

Walasek et al. (2021), Walasek et al. (2022) find that the timing of when sensitive periods emerge depends on the reliability of cues increases across ontogeny, and their results suggest that both information and phenotypic states are modeled, with incremental development in fluctuating environments favouring residual development across ontogeny.

Further, Walasek et al (2024) review current theory and empirical literature and suggest future models could better integrate with empirical research by incorporating phenomena found in empirical systems into our models, and design more models that can be tailored to the life histories of specific species. They encourage for instance that models that incorporate the life history trade-offs we see in the development of alternative phenotypes in bulb mites (between scramblers and fighters) (Smallegange, 2023; Smallegange et al., 2019), and it happens that authors interested in these taxa are also interested in bridging the gap in research between demographic modelling and agency with respect to development shaping demography (Smallegange).

1.2 Developmental windows and sensitive periods

1.3 Demography, agency and life-history

Feedback between plastic responses and population dynamics has been explored for fixed reaction norms by Brass et al. (2021) who showed that ___. However, it is unclear how these eco-devo dynamics will play out over evolutionary time.

1.4 Hypothesis

1.5 Methods

2 The Model Outline

We model a large population of asexual individuals, distributed in a metapopulation across habitat patches of equal size and quality, but each with a population density state with transition probabilities determined by the local population's growth rate. Each individual

has three state dimensions—size (s), patch information (π), and reproductive quality (m)—upon which their developmental strategy is dependent, $x(s, \pi, m)$, which determines the fraction of ingested energy allocated to developing greater reproductive quality over growth to increase their size state (and so ultimately the local population growth rate). Size determines whether or not an individual is mature given they can only reproduce when they reach their maximum size S. An individual's patch information state gives them the subjective probabilities regarding each possible population density state of their habitat patch. Finally, an individual's reproductive quality state m determines their rate of reproduction at maturity.

We assume mutations are rare and of small effect such that effectively all individuals across the metapopulation have the same state-dependent developmental strategy (henceforth referred to as the resident strategy). Given the resident strategy we then do a forward projection using a Markov chain until demographic stability is reached at the metapopulation level (i.e. among patches). Given the probability distribution of population density states of patches, and assuming individuals have complete information as to this probability distribution and of the transition probabilities between patch pop. density states, we find the best response strategy for an invading individual (i.e. the developmental strategy that maximises their expected fitness).

2.1 The Life Cycle and Demographic Feedback

We model the life cycle of both individuals and of the patches they reside in (see figure 1).

1. Newborns disperse into random patches, with accurate prior information as to the probability distribution over patch population density states, and causing transitions in patch population density states.

2. Individuals forage, with food availability being a random variable, normally distributed, with mean determined by the population density.

3. Individuals update their patch information state given informativeness of food availability.

4. Immature individuals allocate a proportion of their ingested energy towards increasing their future reproductive quality given their developmental strategy, the rest of the energy goes towards maturation (increasing their size). Mature individuals do not grow, investing all ingested energy towards current reproduction.

5. Individuals die given a baseline probability of death from exogenous factors and given perturbations in patch quality (e.g. disease outbreaks) influenced by the patch

population density state, and in turn causing transitions of patch states to lower population densities.

6. Individuals that are of maximum size are mature and reproduce at rate determined by their reproductive quality state.

2.2 State dimensions

- Patch population density state, $n \in \{0, 1\}$, corresponding to low and high population density states respectively. We assume that n determines the probability of patch quality decline and determining the availability of food.
- Size state, $s \in \{1, 2, ..., S\}$, where S is their size at maturity.
- Patch belief state, $\pi = \{0, 1/I, 2/I, ..., (I-1)/I, I/I\}$, is an individual's implicit and subjective probability that the patch population density state is high, where L determines the resolution of the belief state.
- Reproductive quality state, $m \in \{0, 1/M, 2/M, ..., M/M\}$, corresponding to the quality of an individual's physiology and so the mass of their offspring. Further, any investment in quality above the quality threshold m_{switch}, results in an individual developing an alternative fighter phenotype, giving additional reproductive benefit $r_{fighter}$.

Part II
Appendix: The Dynamic Game

3 Submodels

Here we describe pseudocode for calculating the demographically stable probability distribution of population densities across patches and of the transition probabilities between patch density states. Patches have a population size state $n \in \{0, 1\}$. The actual population size for any given state q_n is defined as

$$q_n = n_{min} \cdot (n + 1) \tag{1}$$

where n_{min} is the maximum population size of any patch.

To initialize the model we assume that in the first instance the proportion of patches that are of density $q_1$is $\eta = 0.5$, that all individuals are of prior belief state $\pi = \eta = 0.5$ regarding the patch that they are in, and that they are of size state $s = 0$ and of quality state $m = 0$.

3.1 Foraging

Individuals observe cues for population density in the form of food availability. Food availability is a random variable influenced by the population density, all patches produce the same amount of food which must be shared, so for any given patch of population density q_n individuals receive a food mass of

$$g \sim \text{Normal}(\frac{\bar{g}}{q_n}, \sigma_g) \tag{2}$$

where parameter $\bar{g}$ is the mean food mass production of any patch and parameter σ_g is the food mass standard deviation. However, as we are working with discrete mathematics, we must approximate this normal distribution using a pseudo-normal probability distribution over a grid of possible food mass values $\boldsymbol{G} = \{0, 1, ..., G_{max}\}$, with likelihoods for each realised food mass being

$$L(g|q_n) = \tfrac{1}{K} e^{-\frac{(g-\frac{\bar{g}}{q_n})^2}{2\sigma_g^2}} \quad \text{for} \quad g \in \boldsymbol{G}, \tag{3}$$

where N is a suitable integer that determines the fineness of the grid of values, and K is

$$K = \sum_{-G}^{G} e^{-\frac{(g-\frac{\bar{g}}{q_n})^2}{2\sigma_g^2}}. \tag{4}$$

3.1.1 Observation

Given the food mass g, individuals observe the resulting nutrient data and update their population density information states (i.e. their belief states). Assuming Bayesian optimal updating, their posterior belief is defined $\forall i$ as

$$\pi' = \frac{\pi \cdot L(g|q_1)}{\pi \cdot L(g|q_1) + (1-\pi) \cdot L(g|q_0)}. \tag{5}$$

The posterior belief state π' then needs to be interpolated onto a discrete grid of possible values of π which (because it represent a probability distribution) lies on a unit simplex (i.e.

all values are positive and they add up to 1), requiring a grid of barycentric co-ordinates. To this end, the first step is to decompose each π' into it's integer and fractional parts, with the integer part defined by

$$i = \lfloor \pi' \cdot I \rfloor,$$

and subsequently the fractional part defined as

$$f = \pi' \cdot I - i.$$

This then gives the transition probabilities between discrete belief states

$$P(\pi' = i|\pi, g) = 1 - f,$$

$$P(\pi' = i + 1|\pi, g) = f.$$

Finally, for all other possible posterior co-ordinates on the belief grid, the transition probability is

$$P(\pi'|\pi, g) = 0.$$

3.1.2 Development

Each individual, choose a developmental action given their state $\{s, \pi, m\}$, given the resident developmental strategy for that state $x(s, \pi', m) \in (0, G)$. Their developmental action determines the threshold mass above which energy foraged goes to the development of phenotypes of greater reproductive quality (e.g. fighter morphology), generating a transition in their size state s and their reproductive quality state m. Of the portion of energy allocated to quality, all energy below a size-dependent investment threshold goes to maintenance costs. Therefore, if investment in quality falls below this threshold, given their state-dependent strategy and their food mass, there is a discounted probability of survival

$$P(survival|g, s, \pi', m) = min(1, \frac{g - \Delta m - \Delta s}{y(s, m)}) \tag{6}$$

such that if the individual has no energy to allocate to maintenance they die, where $y(s)$ is the size dependent maintenance cost

$$y(s, m) = \frac{s^3 \cdot m}{S^3 \cdot M} \cdot y_{max} \tag{7}$$

where we assume maintenance costs scale by the volume of the individual and by the quality of their somatic cells and developmental systems (given their quality state m), and where y_{max} is the maximum maintenance investment.

The resulting transition probabilities, $p(s', \pi', m'|s, \pi, m)$, between state $\{s', \pi', m'\}$ and state $\{s, \pi, m\}$ given the resident developmental strategy $x(s, \pi, m)$ and discounted survival probabilities, are found be doing the following:

1. For all $\{s, \pi, m, g\}$
$$P(s', m'|s, \pi', m, g) = 0.$$

2. For all g and for each state, calculate the changes in quality and size
$$\Delta m = max(g - x(s, \pi', m), 0) \tag{8}$$
where Δm is the change in quality state of individuals after ingesting a food mass of g, and where
$$\Delta s = max(g - \Delta m - y(s, m), 0), \tag{9}$$
is the change in size state of individuals after ingesting a food mass of g, given that all energy allocated to growth below the fixed and state-dependent maintenance cost threshold $y(s, m)$, below which all food mass is invested in maintenance.

3. These changes then need to be interpolated onto the size and quality grids, and for all interpolated values we find the resulting discretised transitions and their respective probabilities
$$P(s + \Delta s, m + \Delta m|x, s, \pi', m, g) = P(\Delta s) \cdot P(\Delta m)$$

3.2 Demographic stochasticity

Every patch undergoes stochastic perturbations in population density, influenced by patch population density (e.g. the local extinction of a prey species). We assume that the probability of death is equal for all individuals so there is no change in the distribution of individual states within patches as a result of these perturbations. We define the resulting transition probabilities as
$$P_{stoch}(q_0|q_0) = 1 - \gamma_{low}$$
$$P_{stoch}(q_1|q_0) = \gamma_{low}$$
$$P_{stoch}(q_0|q_1) = \gamma_{high}$$
$$P_{stoch}(q_1|q_1) = 1 - \gamma_{high}$$

where the parameters γ_{low} and γ_{high} give the rates of transition up from low patch density states and down from high density states respectively. We assume $\gamma_{high} > \gamma_{low}$, as high density patches are more susceptible to famine and disease outbreaks.

3.3 Reproduction

Adults, individuals of size state $s = S$, reproduce asexually at a rate dependent on their reproductive quality state. We here define $R(s, m, n)$, the expected reproductive rate of an individual of state $\{s, m\}$ in a patch of density q_n, as

$$R(s,m,n) = \begin{cases} \text{if } s \geq S \text{ and } m \geq m_{switch}: & \sum_g P(g|q_n) \cdot g \cdot (r_b + m \cdot r_m + r_{fighter}) \\ \text{if } s \geq S \text{ and } m < m_{switch} & \sum_g P(g|q_n) \cdot g \cdot (r_b + m \cdot r_m) \\ \text{else}: & 0 \end{cases} \tag{10}$$

where S is their size at maturity, m_{switch} is the threshold investment in quality above which the fighter polyphenism is expressed, r_b is the baseline reproductive rate, r_m is the reproductive benefit of greater somatic quality, and $r_{fighter}$ is reproductive benefit of expressing the fighter polymorphism allowing them to extract auxiliary resources for reproduction. To make solving the model tractable we must assume the fighter phenotype does not influence competition for the shared limited resource represented by g.

3.4 Dispersal

3.4.1 Migrant Physiology

Ideally, individuals would be born with precise and accurate prior information regarding the probability distribution over the population density states of patches. Individuals would then update their information state given cues for the population density of their patch which come from the availability of food, see section 1.2. Individuals are born at size state 0 and reproductive quality state 0, and a belief state over population densities of

$$\pi \approx \eta_1 \tag{11}$$

Where η_1 is the proportion of patches that are of density state $n = 1$, and $\eta_0 = 1 - \eta_1$ is the proportion of patches that are of density state $n = 0$. However, to form a discrete grid of states that we can solve a Markov decision process for we need interpolate each π onto a grid of possible discrete probability values, $\pi = i \in \{0/I, 1/I, 2/I, ..., I/I\}$, where I is an integer determining the fineness of the grid. The first step of interpolation is to find the floor of $\eta_1$when it is scaled by the grid size, which will give us the nearest grid cell i below the continuous value

$$i = \lfloor \eta_1 \cdot I \rfloor,$$

and subsequently the difference is the fractional part f defined as

$$f = \eta_1 \cdot I - i.$$

We can then use the fractional part as a probability that the state should be interpolated to be $(i+1)$, finally giving us two discrete states and associated probabilities for defining the state distribution of migrants:

$$\tilde{\rho}(s=1,\pi=i,m=0)=1-f,$$
$$\tilde{\rho}(s=1,\pi=i+1,m=0)=f$$

3.4.2 Patch Densities transition probabilities

Assuming a dispersal rate of μ and a fixed number of patches N, the number of individuals arriving in each patch of population density q_n is

$$\alpha(q_n)=(1-\mu)\cdot q_n\cdot r(q_n)+\frac{\mu\cdot\cancel{N}\cdot\sum_o[\eta_o\cdot q_o\cdot r(q_o)]}{\cancel{N}} \tag{12}$$

where $r(q_n)$ is the expected reproductive rate of an individual in a patch of population density q_n and where η is the proportion of patches that are of density q_n. Note that N in the numerator of the fraction is there to scale the number of migrants given the number of patches producing them, but that N gets canceled out as these migrants are dispersed across these patches (see the denominator), telling us that the number of immigrants to a patch is not dependent on the number of patches; only the frequency distribution over patch states matters. The expected reproductive rate of an individual in a patch of population density q_n is defined as

$$r(q_n)=\sum_s^S\sum_\pi^\Pi\sum_m^M[\rho_n(s,\pi,m)\cdot R(s,m,n)] \tag{13}$$

where $\rho_n(s,\pi,m)$ is the probability that an individual will be of state $\{s,\pi,m\}$ given that they are in a patch of population density q_n, and $R(s,m,n)$ is the reproductive rate of an individual of state $\{s,m\}$ in a patch of density q_n (see section 1.4).

4 Demographic projection and stability

4.0.1 Deterministic deaths and births

Given the baseline rate of death λ and deaths caused by lack of investment in maintenance, the posterior proportions of each physiological state in sub-populations of each patch state n is

$$\hat{\rho_n}(s',\pi',m')=\sum_g\sum_s\sum_\pi\sum_m P(s',m'|s,\pi',m,g)\cdot P(survival|g,s,\pi',m)\cdot P(\pi'|\pi,g)\cdot L(g|q_n)\cdot\rho_n(s,\pi,m) \tag{14}$$

Where, as a reminder, $\rho_n(s, \pi, m)$ is the prior probability distribution over states given the patch is of population size q_n, $P(g|q_n)$ is the probability of a food mass of g in patches of size q_n, $P(\pi'|\pi, g)$ is the probability of transitioning to posterior information state π' given this food mass, $P(survival|s, \pi', m, g)$ is the probability of survival of maintenance costs given their developmental strategy dependent on posterior information state π', and $P(s', m'|s, \pi', m, g)$ is the probability of transitioning to size state s' and quality state m' given the outcomes of development and given that they have survived maintenance costs.

Given the baseline rate of death λ and deaths caused by lack of investment in maintenance, patches can transition to lower population density states. Therefore, given the posterior densities of individuals within patches, the posterior of population size of a patch of density state q_o is

$$u(q_o) = q_o \cdot (1 - \lambda) \cdot \sum_s \sum_\pi \sum_m \hat{\rho}_o(s', \pi', m')$$

The following function returns the posterior population size of a patch of prior density o, prior physiological state distribution $\rho_o(s, \pi, m)$, and resident strategy $x(s, \pi, m)$, given the above birth and death processes

$$n(q_o, \rho_o, x) = u(q_o) + \alpha(q_o).$$

This deterministic update of size then needs to be interpolated onto the grid of possible patch density states giving transition probabilities of a patch from state o to n given deaths from maintenance failures, defined by interpolating $u(o)$ thus for each $o \in \{0, 1\}$

$$P_{\text{det}}(q_1|q_o) = \frac{min(q_1, max(q_0, n(q_o, \rho_o, x))) - n_{low}}{n_{max}} \tag{15}$$

and then

$$P_{\text{det}}(q_0|q_o) = 1 - P_{\text{det}}(q_1|q_o)$$

We then wish to add some demographic stochasticity to this deaths and births process for which we use our model of density-dependent perturbations for each possible transition

$$P(q_n|q_o) = P_{det}(q_n|q_1) \cdot P_{stoch}(q_1|q_o) + P_{det}(q_n|q_0) \cdot P_{stoch}(q_0|q_o)$$

We can then calculate the posterior distribution of patch states

$$\eta_1' = \eta_1 \cdot P(q_1|q_1) + (1 - \eta_1) \cdot P(q_1|q_0), \tag{16}$$

and then the posterior probability distribution over individual states for patches of each population density state

$$\rho_n'(s', \pi', m') = \sum_{q_o} \frac{1}{\eta_n'} \cdot \eta_o \cdot P(q_n|q_o) \cdot [\frac{u(q_o)}{u(q_o) + \alpha(q_o)} \cdot \frac{\hat{\rho}_o(s', \pi', m')}{\sum_s \sum_\pi \sum_m \hat{\rho}_o(s', \pi', m')} + \frac{\alpha(q_o)}{u(q_o) + \alpha(q_o)} \cdot \tilde{\rho}(s, \pi, m)]. \tag{17}$$

4.1 Belief updating between growth cycles

The population evolves priors for the transition probabilities between growth cycles, but in the first instance there will not yet be any evolutionary history for us to calculate these probabilities from and so we will assume that individuals assume patch states remain the same, giving a probability of belief state transition between cycles of $P(\pi' = \pi|\pi, t) = 1$ and so no transition in the state distributions of patches, $\rho''_n(s, \pi, m) = \rho'_n(s, \pi, m)$.

Only after their first iteration of adaptation, described in section 4, will they evolve priors for the transition probabilities between patch population density states at demographic stability given the developmental actions of the rest of the population, patch perturbations, and patch growth from dispersal, since their own previous development action. The transition probability at demographic stability of a patch of population density j to a density of i, given perturbations and dispersal is full therefore, the posterior belief π' becomes

$$\pi' = (1 - \pi) \cdot P(q_1|q_0) + \pi \cdot P(q_1|q_1)$$

which then has to be interpolated on the discrete belief state space as before, with the integer part defined by

$$i = \lfloor \pi' \cdot I \rfloor,$$

and subsequently the fractional part defined as

$$f = \pi' \cdot I - i.$$

We can then generate the transition probability matrix for belief states given that population growth and patch perturbations will have occurred within the time between developmental actions, i.e. $p(\pi'|\pi, time)$, defined by running the following algorithm for all prior belief states:

1. $\forall \pi'$, $P(\pi'|\pi, time) = 0$
2. add $(1 - f)$ to $P(\pi' = i|\pi, time)$,
3. add f to $P(\pi' = i + 1|\pi, time)$.

and the distribution of states must again be updated

- add f to $\rho''_n(s, i + 1, m)$
- add $(1 - f)$ to $\rho''_n(s, i, m)$

4.2 Demographic Stability

Assuming that with successive generations n that the distribution of patches over states converges on stable values, i.e.

$$\eta_1^* = \lim_{\to\infty}[\eta_1] \tag{18}$$

further, we can approximate η_1^* by iterating over generations until

$$tol > |\eta_1 - \eta_1''| \tag{19}$$

where the tolerance threshold, tol, is very small, and η_1'' is the proportion of patches that are of population density state q_1 after dispersal and deaths. Once demographic stability is reached we assume $\eta_1^* = \eta_1$ and proceed with invasion analysis.

5 Adaptation and Evolutionary Stability

Having found the demographically stable state of the metapopulation, given the resident strategy, we can then use the Bellman equation to find the best response strategy $b(x(s, \pi, m))$ for an invading individual. We then assume the best response strategy becomes the new resident strategy, and recalculate the demographically stable state of the metapopulation. The evolutionarily stable strategy (ESS) is a resident strategy for which invading individuals can do no better than to follow the resident strategy and so the ESS is defined as

$$\bar{x}(s, \pi, m) = b(x(s, \pi, m)) = x(s, \pi, m). \tag{20}$$

Assuming there is a singular ESSs, we can iterating over invasions till the best response strategy converges on the ESS, and the starting point of the population will not matter. By sampling from multiple starting points we can test this assumption, and if necessary plot out the adaptive dynamics with phase portraits where there are multiple ESSs or oscillatory dynamics.

To find the best response we define a Bellman fitness equation for individuals in the model where $V'(s, \pi, m) = 0$ for all state combinations (as the individual is dead), and iterate backwards to the previous time step

$$V'(s, \pi, m) = R(s, m) + \max_{x(s,\pi,m)=\kappa}[\sum_{s'}\sum_{\pi''}\sum_{m'} a(s', \pi'', m'|s, \pi, m, \kappa) \cdot V(s', \pi'', m')], \tag{21}$$

where $R(s, m)$ is the rate of reproduction for an individual of size s and quality m (see equation 1), $p_{pert}(survival|\pi_n)$ is the probability the individual survives between time-steps given patch quality perturbations, and $p(s', \pi'', m', survives|s, \pi, m, \kappa)$ is the survival (given maintenance) and state transition probability to the posterior state $\{s', \pi'', m'\}$ of individuals of prior state $\{s, \pi, m\}$.

5.1 State transitions between developmental actions

Individuals are born with accurate prior information regarding the probability distribution over the population density states of patches

$$\pi = \eta_1 \tag{22}$$

Individuals also know the transition probabilities between patch population density states at demographic stability given the developmental actions of the rest of the population, patch perturbations, and patch growth from dispersal, since their own previous development action. The transition probability at demographic stability of a patch of population density j to a density of i, given perturbations and dispersal is full therefore, the posterior belief π' becomes

$$\pi' = (1 - \pi) \cdot \tilde{P}(q_1|q_0) + \pi \cdot \tilde{P}(q_1|q_1) \tag{23}$$

which then has to be interpolated on the discrete belief state space as before, with the integer part defined by

$$i = \lfloor \pi' \cdot I \rfloor,$$

and subsequently the fractional part defined as

$$f = \pi' \cdot I - i.$$

We can then generate the transition probability matrix for belief states given that population growth and patch perturbations will have occurred within the time between developmental actions, i.e. $P(\pi'|\pi, time)$, defined by running the following algorithm for all prior belief states:

1. $\forall \pi'$, $\tilde{P}(\pi'|\pi, time) = 0$
2. add $(1 - f)$ to $\tilde{P}(\pi' = i|\pi, time)$,
3. add f to $\tilde{P}(\pi' = i + 1|\pi, time)$.

The posterior belief π' is then updated again upon receiving a food mass of g for each of $i \in \{0, 1, 2\}$

$$\pi''_i = \frac{\pi' \cdot P(g|q_1)}{\pi' \cdot P(g|q_1) + (1 - \pi') \cdot P(g|q_0)}$$

which then has to be interpolated on the discrete belief state space as before, with the integer part defined by

$$i = \lfloor \pi' \cdot I \rfloor,$$

and subsequently the fractional part defined as

$$f = \pi' \cdot I - i.$$

We can then generate the transition probability matrix for belief states given a food mass of g for each prior belief state π'

1. $\forall \pi''$, $P(\pi''|\pi', g) = 0$
2. add $(1 - f)$ to $P(\pi'' = i|\pi', g)$,
3. add $f$to $P(\pi'' = i + 1|\pi', g)$.

Individuals have a probability of death from patch depletion and a probability of death λ from exogenous factors (e.g. old age). Given an individual's information state after foraging π', and given the probability of stochastic patch density transitions at demographic stability $\tilde{P}_{stoch}(q_k|q_j)$, the probability that an individual dies due to patch depletion is

$$P_{pert}(survival|\pi') = (1 - \lambda) \cdot \sum_j \sum_k \pi'_j \cdot (1 - \tilde{P}_{stoch}(q_k|q_j) \cdot \frac{q_j - q_k}{q_j}) \tag{24}$$

5.2 State Transitions During Development and Perturbations

As observations of patch density and an individual's development are linked by the same random variable, food mass g, we define the state transition probability given individual decision making as

$$a(s', \pi''', m'|s, \pi, m, \kappa) = P_{pert}(survival|\pi') \cdot \sum_{\pi'} [P(\pi'|\pi, time) \cdot \sum_g [P(g|\pi) \cdot P(\pi'|\pi, g) \cdot P(s', m', survives|x \tag{25}$$

where $p(s', m', survives|\kappa, s, m, \pi'', g)$ is the probability of posterior size s' and quality m' given the individual's developmental policy dependent on their current size, quality, and information states, given the food mass they received, and given their probability of survival given maintenance costs, all defined once by doing the following:

1. For all $\{s, \pi, m, g, \kappa\}$
$$P(s', m'|s, \pi, m, g, \kappa) = 0.$$
2. For all i and all g, define the transitions in state predicted by their developmental strategy κ,where
$$\Delta m = max(g - \kappa, 0) \tag{26}$$

is the change in quality state of individuals after ingesting a food mass of g, and where

$$\Delta s = max(g - \Delta m - y(s, m), 0), \tag{27}$$

is the change in size state of individuals after ingesting a food mass of g, given that all energy allocated to growth below the fixed and state-dependent maintenance cost threshold $y(s, m)$, below which all food mass is invested in maintenance.

3. Then add $P(survival|g, s, m, \kappa)$ to $P(s + \Delta s, m + \Delta m, survives|s, \pi, m, g, \kappa)$, where depending on the food mass g an individual encounters and the fraction of that mass that they invest in maintenance, there is a probability of death from lack of maintenance of

$$P(survival|g, s, m, \kappa) = min(1, \frac{g - \Delta m - \Delta s}{y(s, m)}) \tag{28}$$

where $y(s, m)$ is the size and quality dependent maintenance cost threshold to avoid death completely, defined as shown previously:

$$y(s, m) = \frac{s^3 \cdot m}{S^3 \cdot M} \cdot g_{max} \tag{29}$$

6 Life history projection

Now that we have found the state-dependent evolutionarily stable strategy for a given set of parameter values, along with the associated demographically stable distribution of individual physiological states and patch density states, we can ask what the ESS behaviour of a cohort of individuals within such a population would look like. We can assume that one small cohort following the same ESS will not have an influence on the distribution of states of the larger population and so we do not need to update any of the transition probabilities.

We know that they will begin life at the minimum size and quality states and with their belief state being the evolved prior for the stable patch distribution prior to dispersal, η_1^*. Where η_1^* is the proportion of patches prior to dispersal that are of density state $n = 1$, and $\eta_0 = 1 - \eta_1$ is the proportion of patches that are of density state $n = 0$. However, to form a discrete grid of states that we can solve a Markov decision process for we need interpolate each π onto a grid of possible discrete probability values, $\pi = i \in \{0/I, 1/I, 2/I, ..., I/I\}$, where I is an integer determining the fineness of the grid. The first step of interpolation is to find the floor of $\eta_1^*$when it is scaled by the grid size, which will give us the nearest grid cell i below the continuous value

$$i = \lfloor \eta_1 \cdot I \rfloor,$$

and subsequently the difference is the fractional part f defined as

$$f = \eta_1 \cdot I - i.$$

We can then use the fractional part as a probability that the state should be interpolated to be $(i+1)$, finally giving us two discrete states and associated probabilities for defining the state distribution of this cohort as

$$\rho_{i,t}(s, \pi = i, m) = 1 - f$$

$$\rho_{i,t}(s, \pi = i+1, m) = f$$

Very conveniently we already know the transition probabilities, $a(s', \pi'', m'|s, \pi, m, \kappa = \bar{x}(s, \pi, m))$, for individuals using the ESS strategy $\bar{x}(s, \pi, m)$, and the system is both at demographic and evolutionary stability; making demographic projection at this stage exceedingly simple:

$$\rho_{i,t+1}(s', \pi'', m') = \sum_{s} \sum_{\pi} \sum_{m} a(s', \pi'', m'|s, \pi, m, \kappa = \bar{x}(s, \pi, m)) \cdot \rho_{i,t}(s, \pi = i, m) \tag{30}$$